

Chapter 76A

Data Analyst & Data Scientist Question Bank

Answer the role-specific questions a Data Analyst or Data Scientist interview actually asks, distinct from the skill-testing banks elsewhere in this part (SQL, Python, stats, ML) · walk an interviewer through a real project end to end, at the right level of depth for each role · handle the...

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Chapter 76A. Data Analyst & Data Scientist Question Bank

Part VIII — The Interview Playbook

A note on this chapter's numbering. Like Chapter 72A, this chapter sits outside the original plan, added on author request as a role-specific companion to Chapter 76B (Business Analyst). It's numbered **76A**, a placeholder for the coordinator to renumber at final assembly, exactly the convention already established for Chapter 72A.

You will learn to: answer the role-specific questions a Data Analyst or Data Scientist interview actually asks, distinct from the skill-testing banks elsewhere in this part (SQL, Python, stats, ML) · walk an interviewer through a real project end to end, at the right level of depth for each role · handle the "what's the difference between a DA and a DS" question, and the harder version, "which one are you actually best suited for" · survive a portfolio deep-dive without getting caught flat-footed on a detail you glossed over.

How this chapter is different from Chapters 70–75. Those chapters test whether you *can do the work* (write SQL, debug pandas, reason about a p-value). This chapter tests whether you can *talk about the work you've done*, structure a project narrative, and demonstrate the specific judgment a DA or DS role requires beyond raw technical skill. Same format throughout: **"Remember it as..."** hooks, tier tables, scan tables for rapid-fire.

76A.1 What the two roles actually do, and how interviewers probe the line between them

Q76A-001 · What's the actual difference between a Data Analyst and a Data Scientist, in a way that isn't just "DS does machine learning"?

Remember it as: A Data Analyst answers questions that already exist. A Data Scientist often has to decide which question is even worth asking, and builds something that keeps answering it automatically.

Answer in one line: A **Data Analyst** primarily investigates specific, often one-off business questions using existing data, producing reports, dashboards, and recommendations for human decision-makers; a **Data Scientist** more often builds systems (models, algorithms) that make or support decisions repeatedly and automatically, and spends more time on problem framing, experimentation, and statistical rigor before any model exists at all.

Tier	What to say
Passes	"DA does reporting, DS does machine learning" (true often enough, misses the deeper distinction)
Strong	The one-off-question vs. repeatable-system distinction above, with a concrete example of each: a DA answering "why did Q3 sales drop in the East region" once, versus a DS building a churn model that scores every customer every day going forward

Tier	What to say
Extra points	+ [Business] the overlap is large and growing, especially at smaller companies where one person does both; naming that honestly, rather than presenting a rigid boundary, signals real experience over a textbook answer + [Edge cases] a DA role at a company without a dedicated DS function often includes real modeling work, and a DS role at a data-mature company can be dominated by dashboarding and ad-hoc analysis, so the title alone doesn't fully predict the day-to-day

Likely follow-ups: Which parts of your own experience map more to one role than the other? Where do you see yourself wanting to grow, toward more DA-style or more DS-style work? **Red flag:** a rigid, textbook boundary with no acknowledgment of real-world overlap, or being unable to place your own experience on either side of it. **Learn it in:** Chapter 7, §7.2 (the ten-role/track map this whole book is built around).

Q76A-002 · An interviewer asks "why do you want this DA role and not a DS role, given your background touches both?" How do you answer honestly without talking yourself out of the job?

Remember it as: This question is testing self-awareness and genuine fit, not loyalty. A specific, honest reason beats a generic "I love both equally."

Answer in one line: Give a specific, honest reason tied to what you actually enjoy or want more of right now (closer to business stakeholders, faster feedback loops, breadth over depth, or the reverse for a DS answer), not a vague "I'm passionate about data" that could apply to either role equally.

Worked example:

"I've done modeling work and enjoyed it, but what energizes me more day to day is the moment right after I hand someone an answer and watch them actually act on it, that feedback loop is faster and more visible in DA work than in a lot of DS work, where the payoff can be months away. That's a real, specific reason I'm choosing this role now, not a consolation prize."

Tier	What to say
Passes	"I like both, but this role sounded interesting" (generic, doesn't actually answer why this one specifically)
Strong	The specific, honest framing above, tied to a real preference, not a rehearsed line
Extra points	+ [Business] naming a genuine trade-off (faster feedback vs. deeper technical depth) rather than pretending one role is strictly better shows the kind of self-aware judgment interviewers are actually screening for here

Likely follow-ups: What would make you consider a DS role again in the future? What's one thing from DS work you'd want to bring into a DA role? **Red flag:** an answer so generic it could be copy-pasted into an interview for the opposite role with no changes. **Learn it in:** Chapter 68, §68.6 (the same self-honest framing used for a 30/60/90-day plan, applied here to a "why this role" answer).

Rapid-fire, 76A.1

#	Question	One-line answer	Extra point
Q76A-003	Where does a Data Analyst's work typically end and a Data Engineer's begin?	A DA typically works with data already made accessible (a warehouse, a clean table); a DE builds and maintains the pipelines that get it there in the first place	[Learn it in] Chapter 7, §7.2's full ten-role map
Q76A-004	Is "Business Intelligence Analyst" the same role as "Data Analyst"?	Largely overlapping, with BI leaning more heavily toward dashboards and reporting tools specifically, and DA sometimes implying more ad-hoc, code-based analysis	[Business] titles vary enormously by company; always check the actual JD's must-have list (Chapter 68, §68.2) rather than assuming from the title alone
Q76A-005	What's the most common reason a DA candidate is passed over for being "too DS," or vice versa?	Over-indexing the interview on technical depth (model internals, algorithm choice) when the role is really about stakeholder-facing analysis, or the reverse: under-demonstrating statistical rigor in a DS interview by staying purely descriptive	[Business] matching your answer's depth and focus to the actual role, not just showcasing everything you know, is itself part of what's being evaluated
Q76A-006	Does a Data Scientist need to know SQL as well as a Data Analyst does?	Yes, in almost every real DS role; the depth of Python/ML skill differs sharply between the two roles, but SQL fluency is close to a shared baseline	[Learn it in] Chapter 71, all of it, applies equally to both roles

76A.2 Walking through a Data Analyst project, end to end

Q76A-007 · "Walk me through a data analysis project you're proud of." Give a strong, structured answer

Remember it as: *Business question first, method second, impact last, and the impact needs a number, even a rough one.*

Answer in one line: Structure the narrative as: the business question and why it mattered, what data and method you used (briefly, not a code walkthrough unless asked), what you found, and critically, what changed as a result, ideally with a quantified impact.

Worked example:

"Riverstone's sales team was manually pulling lead data weekly to decide who to call, taking about three hours every Monday. I built a scored, prioritized call list instead: I pulled the historical conversion data, found that lead source and company size were the strongest predictors of conversion, and built a simple scoring rule the sales lead could act on without needing a model explained to them. It cut their weekly prep time from three hours to about twenty minutes, and conversion on the top-scored leads was measurably higher than the previous first-come-first-served approach, about 12% versus 7%. The part I'm most proud of isn't the analysis itself, it's that I checked back a month later and it was still being used daily, which isn't always true of analysis work."

Tier	What to say
Passes	Describes the technical steps in detail (the query, the tool used) with no mention of the business problem or the outcome
Strong	The business-question-first, quantified-impact structure above
Extra points	+ [Business] the closing line, "I checked back a month later," directly answers the unspoken follow-up every interviewer wants to ask but often doesn't: did this actually stick, or was it a one-time deliverable nobody used again + [Validate] the specific before/after numbers (three hours to twenty minutes, 7% to 12%) give the interviewer something concrete to probe further, rather than a vague "it really helped"

Likely follow-ups: What would you do differently if you started this project again today? What was the hardest part, and how did you get past it? **Red flag:** a project narrative that never states a business impact, or states one with no number attached at all. **Learn it in:** Chapter 69's whole Extra-Points Method, this is the same five-dimension rubric applied to a full project narrative instead of a single technical question.

Rapid-fire, 76A.2

#	Question	One-line answer	Extra point
Q76A-008	An interviewer asks "what was the hardest technical challenge in that project?" What are they actually testing?	Whether you can go one level deeper than the polished summary, showing real problem-solving under a specific constraint, not just reciting the project again	[Business] have one specific, technical sticking point ready for every project in your portfolio, not just the smooth headline version
Q76A-009	How would you handle being asked to present the same project to a technical panel and then to a business stakeholder, back to back?	Keep the core findings identical; adjust the depth of method explanation and vocabulary for each audience, exactly Chapter 24's storytelling discipline	[Learn it in] Chapter 24 (storytelling with data, bottom line up front)
Q76A-010	What if your proudest project doesn't have a clean, positive outcome to report?	Report it honestly, including what you learned from a result that didn't pan out; a thoughtful account of a project that taught you something is often more convincing than a suspiciously perfect success story	[Real evidence] interviewers are often more interested in how you handle an ambiguous or negative result than in a flawless win
Q76A-011	Should you mention tools and technologies by name in a project walkthrough, or focus purely on the business story?	Both, but in order: lead with the business story, mention tools briefly and naturally as part of the "how," not as a checklist recited up front	[Business] a walkthrough that opens with a tool list ("I used Python, pandas, and Tableau") before explaining what problem it solved reads as technical-first, not business-first

76A.3 Walking through a Data Science project, end to end

Q76A-012 · "Walk me through an ML project you've built." Give a strong, structured answer

Remember it as: Problem framing and evaluation choice matter as much as the model itself. A DS project narrative that jumps straight to "I used gradient boosting" skips the parts that actually show judgment.

Answer in one line: Structure the narrative as: the business problem and why it needed a model (not just a rule or a report), how the target and evaluation metric were chosen and why, what approach was tried and what it was compared against, and what happened once it was in front of real users or real data, including any honest limitation.

Worked example:

"The business problem was prioritizing which leads sales should call, and a simple rule wasn't capturing enough of the pattern in the data to be useful. I framed it as a binary classification problem, predicting conversion within 30 days, and I started with logistic regression as a baseline before trying gradient boosting, since I wanted to know whether the added complexity was actually earning its keep, not assume it would. Boosting won by a real margin on held-out data, 0.839 AUC against 0.797. But raw AUC wasn't the full story: I built a cost-based threshold instead of the default 0.5, since a missed high-value lead and a wasted call to a cold one have very different real costs, and that threshold choice is what actually made the model useful to the sales team, not the AUC number alone. The honest limitation: it's a static model right now, retrained monthly, so a sudden shift in lead quality between retrainings wouldn't be caught automatically, and that's the next thing I'd want to build."

Tier	What to say
Passes	Jumps straight to naming the algorithm used, with no mention of problem framing, baseline comparison, or evaluation choice
Strong	The full structure above: framing, baseline-vs-complex comparison, evaluation choice tied to real cost, and an honest limitation
Extra points	+ [Business] naming the cost-based threshold as the thing that "actually made the model useful," not the raw AUC, shows the exact judgment a DS interview is probing for: knowing that a good metric and a useful model aren't automatically the same thing + [Edge cases] volunteering a real limitation (a static, monthly-retrained model) unprompted, rather than waiting to be asked, is itself an extra-point move: it signals honest self-assessment instead of oversell

Likely follow-ups: How would you detect the model going stale between retrainings? What would you have done differently if boosting hadn't beaten the baseline? **Red flag:** a project narrative dominated by algorithm names with no mention of why that algorithm, what it was compared against, or how success was actually measured. **Learn it in:** Chapter 74, all of it (this narrative structure is the same rigor the ML question bank tests, applied to a full project story instead of an isolated question).

Rapid-fire, 76A.3

#	Question	One-line answer	Extra point
Q76A-012B	"What would you do if you had ten times more data for this project?" What's this really testing?	Whether you understand what's actually limiting your current model, more data helps most with variance/overfitting problems, not with a fundamentally wrong feature set or framing	[Learn it in] Chapter 74, Q74-001 (bias-variance)
Q76A-013	How do you decide when a DS project is "done" versus needing more iteration?	When the marginal improvement from further work no longer justifies its cost against the business need, not when the model can't technically be improved any further at all	[Business] "done" is a business judgment call as much as a technical one; naming that explicitly is itself a strong answer
Q76A-014	An interviewer asks you to defend a modeling choice you'd actually reconsider today. How do you answer honestly?	Say so directly: explain the original reasoning, then state plainly what you'd do differently now and why, rather than defending a choice you no longer believe in	[Learn it in] Chapter 69's Move 10 (admit limits honestly)
Q76A-015	Should a DS candidate be able to explain their model to a non-technical stakeholder, or is that a DA-specific skill?	No, it's expected of both; a DS who can't translate model output into a decision a business person can act on is significantly less valuable regardless of model quality	[Learn it in] Chapter 39, §39.7's SHAP-based explanation discipline, built exactly for this

76A.4 Stakeholder communication for DA and DS specifically

Q76A-016 · A stakeholder asks you to make a dashboard "simpler" with no further detail. As a DA, how do you respond, live?

Remember it as: "Simpler" almost always means "I can't find the thing I actually need," not "there's too much visual complexity."

Answer in one line: Ask what they're specifically trying to find or decide when they open it, since "simpler" is usually a proxy for "I can't quickly find what I need," a findability problem, not necessarily a raw information-density problem.

Worked example:

"I'd ask: 'When you say simpler, is it that there's too much on the page, or that it's hard to find the one number you actually check first?' Those need very different fixes. If it's the second, reorganizing what's prominent versus what's secondary might solve it without removing any real information at all."

Tier	What to say
Passes	Immediately starts removing charts and metrics from the dashboard without asking what's actually driving the complaint
Strong	The diagnostic question above, distinguishing a findability problem from a genuine information-overload one

Tier	What to say
Extra points	+ [Learn it in] this is the identical diagnostic instinct as Chapter 76B's Q76-001 ("make the report better"), the same underlying discipline: a vague complaint needs one specific clarifying question before any change is made

Likely follow-ups: What if simplifying for this stakeholder makes the dashboard worse for a different stakeholder who uses the same one? **Red flag:** treating "simpler" as a literal instruction to delete content, with no diagnosis of the real underlying complaint. **Learn it in:** Chapter 76B, Q76-001 and Chapter 70, §70.11's dashboard-critique discipline.

Rapid-fire, 76A.4

#	Question	One-line answer	Extra point
Q76A-017	A DS candidate is asked to explain a model's prediction to someone who's never heard of a p-value. What's the winning move?	Skip statistical vocabulary entirely; explain in terms of the specific factors driving <i>this</i> prediction, using SHAP-style "this account scored high risk because of X and Y," concrete and case-specific	[Learn it in] Chapter 39, §39.7
Q76A-018	How would a DA and a DS each respond differently to "can you just change the number"?	The underlying discipline (diagnose before responding) is identical; a DA more often faces this on a reporting number, a DS more often faces a variant of it as "can you make the model say what we expect"	[Learn it in] Chapter 76B, Q76-033
Q76A-019	What's the risk of a DS presenting only the model's strongest result and omitting a weaker segment's performance?	It's a form of the same overclaiming problem Chapter 39 warns against; a stakeholder who later discovers the omitted weak segment loses trust in every future result, not just that one	[Learn it in] Chapter 39, §39.8 (fairness/segment checks)
Q76A-020	How would you push back on a stakeholder who wants a specific chart type you think is misleading for their data?	Show, don't just tell: build the alternative chart alongside the requested one and let the clearer version make the case, rather than only arguing abstractly against the original request	[Learn it in] Chapter 70, §70.11's dashboard critiques, the same "here's specifically why this distorts the data" discipline

76A.5 Portfolio deep-dives and handling scrutiny

Q76A-021 · An interviewer picks one number from your resume ("you say this improved conversion by 12%") and asks you to defend it in detail. What do they actually want?

Remember it as: A resume number is a claim. The interviewer is checking whether you actually understand how you measured it, not whether the number itself was impressive.

Answer in one line: They want to know exactly how that number was measured (against what baseline, over what time period, with what confounders considered), since a candidate who can't explain their own headline metric in detail either didn't do the analysis rigorously or is inflating the

claim.

Worked example:

"That 12% was conversion on leads flagged by the scoring model compared to the prior quarter's average conversion rate for leads worked in the old first-come-first-served order. I did check that it wasn't just a seasonal effect, the prior quarter's comparison period had similar seasonality, and I also checked that the sales team's overall headcount and effort level hadn't changed materially in that window, since a confound there would have inflated the number without the model actually being responsible."

Tier	What to say
Passes	Can restate the number but can't explain the comparison baseline or any confounders considered
Strong	The full explanation above: baseline, time period, and at least one confound actively checked and ruled out
Extra points	+ [Business] proactively naming a confound you checked and ruled out (seasonality, headcount) is a stronger answer than waiting for the interviewer to ask "but could it have been X instead," since it shows the rigor was applied before the metric was ever put on a resume

Likely follow-ups: How would you know if that 12% wasn't statistically significant, just noise? What would make you distrust your own number here? **Red flag:** unable to state the exact baseline a resume metric was measured against, or visible defensiveness rather than genuine engagement with the scrutiny. **Learn it in:** Chapter 73, all of it (statistical rigor is exactly what's being probed here) and Chapter 69's Move 7 (validate).

Rapid-fire, 76A.5

#	Question	One-line answer	Extra point
Q76A-022	An interviewer finds a small inconsistency between two things you said about the same project. How do you handle it live?	Acknowledge it directly and clarify which version is accurate, rather than doubling down or getting flustered; a small, honestly-corrected inconsistency is far less damaging than visible defensiveness about it	[Learn it in] Chapter 69's Move 10
Q76A-023	What's the risk of over-preparing a polished, memorized project narrative?	It can sound rehearsed and brittle under a genuine follow-up question that deviates from the memorized script, revealing the narrative was recited rather than truly understood	[Business] know the underlying facts cold; don't memorize the exact wording
Q76A-024	An interviewer asks about a project on your resume from three years ago, in specific detail. How do you handle gaps in your memory honestly?	Say plainly what you remember clearly and what's genuinely fuzzy after this long, rather than fabricating confident-sounding detail you don't actually have	[Red flag] confidently inventing specifics for a memory gap is far more damaging if caught than an honest "I don't recall that exact detail, but here's what I do remember"

#	Question	One-line answer	Extra point
Q76A-025	Should you bring a laptop with your actual project code to an interview, in case you're asked to show it?	Generally yes if the interview is portfolio-focused and it's practical; being able to show real code and real output beats describing it from memory alone	[Business] if you do, make sure it actually runs before the interview, nothing undercuts credibility faster than a live demo failing

76A.6 Live-coding-style walk-throughs

Q76A-026 · Full scenario, talked through live: an interviewer says "your resume says you 'used machine learning to improve retention.' What does that actually mean?"

What they're really testing: whether a vague, resume-friendly phrase can be unpacked into the real, specific work behind it, live, without stalling.

Talked through live, start to finish:

"Fair to push on that, it's a compressed phrase covering a few specific things. Concretely: I built a churn prediction model, gradient boosting, trained on twelve months of account activity data, predicting churn within the next 90 days. The 'improve retention' part specifically means: I turned that model's output into a monthly call list for the sales team, ranked by expected revenue at risk rather than churn probability alone, since a large account with a modest risk score is worth more attention than a small account with a high one. What I can't fully claim is that retention definitively improved *because of* the model; what I can say is the accounts on the list were called, and a good share of them didn't churn, but proving causation would need a proper holdout test I haven't run yet, and I'd say that honestly if asked."

Extra-points moves demonstrated: **[Structure]** unpacked the vague phrase into its specific, real components rather than restating it more confidently. **[Business]** distinguished expected-value ranking from probability-alone ranking, exactly Chapter 44's own capstone lesson. **[Real evidence]** stated the actual causal limitation honestly rather than overclaiming the resume phrase's implied certainty.

Likely follow-ups: How would you design that holdout test to actually prove causation? What would you change about how you'd phrase this on a resume, now that you've unpacked it? **Red flag:** doubling down on the vague resume phrase instead of engaging with the specific unpacking the question is asking for. **Learn it in:** Chapter 44 (the capstone project this exact example draws from) and Chapter 73, §73.4 (what a real causal test would need to look like).

Common mistakes and how to spot them

Mistake	Symptom	Fix
A project narrative that opens with tools, not the business problem	Interviewer struggles to follow why the work mattered	Business question first, method second, impact last
No quantified impact in a project walkthrough	A vague "it really helped the team" with nothing to probe further	Always have at least a rough number, honestly caveated if it's rough
A rigid DA-vs-DS boundary with no acknowledgment of overlap	Sounds like a textbook definition, not real experience	Name where your own work actually sits, including the overlap
Defending a resume metric with no memory of its actual baseline	Signals the number may have been inflated or poorly measured	Know exactly how every headline number on your resume was calculated
A DS narrative dominated by algorithm names	Misses the framing, evaluation, and judgment an interviewer actually wants to see	Lead with problem framing and evaluation choice, not the model name
Doubling down when caught in a small inconsistency	Reads as defensiveness, damages trust more than the original slip	Acknowledge it plainly and move on
A memorized, over-polished project script	Breaks under a genuine follow-up that deviates from the script	Know the facts; don't memorize the wording

In the real world: the resume line that became the whole interview

Kavita, interviewing for a Data Analyst role after a career transition, has one line on her resume: "Automated a weekly report, saving the team 3 hours a week." The interviewer spends nearly ten minutes on that single line, not because it's the most impressive thing on the page, but because it's specific and checkable, and specific claims are exactly what a good interviewer knows how to probe.

She's asked: which report, what was manual about it before, how exactly was "3 hours" measured, did anyone verify the automated version was actually correct before it replaced the manual one, and what happened the one time it broke. Because she'd genuinely done the work and understood it, not just summarized it for a resume, she has a real, specific answer to each question, including an honest one about the time it broke: a source file's format changed upstream without warning, and she'd since added a validation check specifically because of that failure.

The interviewer's note afterward: *"The resume line was modest. The depth behind it was not. Ten minutes on one bullet point told us more than the rest of the resume combined."* The lesson generalizes: every single claim on a resume should be able to survive exactly this kind of sustained, specific scrutiny, not because interviewers are trying to catch you out, but because depth on a real claim is one of the fastest, most reliable signals of real work versus a polished summary of someone else's.

Tools

No new tools beyond what earlier chapters in this part already cover. The one habit worth building specifically for this chapter: after finishing any real project, write a five-line summary immediately (business question, method, finding, impact number, one honest limitation) while it's fresh, so months later in an interview you're recalling a summary you already wrote carefully once, not reconstructing the whole thing from memory under pressure.

The project

Goal: prepare two full project narratives, one DA-style and one DS-style (or two of whichever type matches your own background), to the standard this chapter models.

1. For each project, write the five-line summary described above: business question, method, finding, quantified impact, honest limitation.
 2. Practice defending your single most impressive resume metric out loud: state its exact baseline, time period, and one confound you checked.
 3. Write your honest, specific answer to "why this role and not the other one," using Q76A-002's structure.
 4. Have someone else interrupt your project walkthrough with an unexpected follow-up question, and practice answering it without falling back on a memorized script.
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Final-week revision list

Q76A-001, Q76A-002, Q76A-007, Q76A-012, Q76A-016, Q76A-021, Q76A-026.

Key terms

Data Analyst vs. Data Scientist (role distinction) · business-question-first narrative structure · quantified impact · baseline (for a claimed metric) · confound (in a resume claim) · problem framing (DS) · cost-based threshold (referenced) · honest limitation · portfolio deep-dive · resume scrutiny

Where this leads

- **Chapter 69, The Extra-Points Method**, is the rubric and move set every answer above is written against.

- **Chapter 76B, Business Analyst Question Bank**, covers the same "diagnose before responding" and stakeholder-pushback discipline from the BA angle.
- **Chapter 44** (this book's own ML capstone) is the real source for this chapter's worked DS project example, kept consistent with that earlier chapter rather than invented fresh.
- **Chapter 81, Behavioral, HR & Offer Conversations**, is where this chapter's project-narrative skills extend into full STAR-format behavioral stories.